

Lecture 7

More on Entanglement

Hilbert's Place, summer 1935:

Two scruffy regulars come through the swinging doors, in the midst of an intense conversation. The one with the wild grayish hair and frayed sweater says, "No, I will not accept your theory unless you can tell me what the elements of physical reality are."

The other one looks around, throws up his hands in obvious frustration, and says to Art and Lenny, "There he goes again. Elements of physical reality, EPRs, EPRs, that's all he ever thinks about. Albert, stop being obsessive and just accept the facts."

"Never! I cannot accept that one can know everything there is to know about a thing, and still know nothing about its parts. That's utter nonsense, Niels."

"Sorry, Albert. That's just the way it is. Here, let me buy you a beer."

In this lecture, we will look at entanglement in greater depth. To do that, we'll need some additional mathematical tools. First, we'll find out how to work with tensor products in component form. Then, we'll learn about a new operator called the *density matrix*. These tools are not inherently hard to